

Power Running Board (PRB)



Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

The Power Running Boards (PRBs) are controlled by the PRBs module. The PRB motors receive motor power and ground from the PRB module. The PRB module receives door ajar switch status and vehicle speed information from the Controller Area Network (CAN) .

When in AUTO mode, the PRBs will deploy when a door on the same side of the vehicle is opened. When both doors on the same side are closed, the PRBs will retract.

When in OUT mode, both PRBs will stay deployed until OUT mode is deselected, reset to AUTO mode or vehicle speed is greater than 8 km/h (5 mph). Mode will automatically switch back to AUTO once above 8 km/h (5 mph).

When an obstacle is detected, the PRB will reverse direction. If a second obstacle is detected, the PRB will reverse direction again. If a third obstacle is detected, the PRB will stop against the third obstacle until the following conditions are met:

- Change in door ajar status
- Change in the message center status (AUTO, OUT, OFF)
- Vehicle speed greater than 8 km/h (5 mph)

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Power Running Board (PRB)	• Battery Junction Box (BJB) fuse 11 (30A)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00 to diagnose no response from the PRB module.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record Continuous Memory Diagnostic Trouble Codes (CMDTCs) .

8. Clear the CMDTCs and carry out the self-test diagnostics for the PRB module.

9. If the DTCs retrieved are related to the concern, go to Power Running Board (PRB) Module DTC Chart. For all other DTCs, refer to Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Chart

Power Running Board (PRB) Module DTC Chart

DTC	Description	Action
B1162	Power Running Board Hall Effect Sensor Power Circuit Short to Battery	GO to Pinpoint Test E.
B1163	Left Power Running Board Hall Effect Sensor Failure	GO to Pinpoint Test E.
B1164	Power Running Board Hall Effect Sensor Power Circuit Short To Ground	GO to Pinpoint Test E.
B1167	Left Power Running Board Deploy and/or Stow Circuit Short To Battery	GO to Pinpoint Test D.
B1168	Left Power Running Board Stow Circuit Short To Ground	GO to Pinpoint Test D.
B1169	LH Motor Traveled Too Far	LH motor is freewheeling (broken geartrain linkage). INSTALL a new LH Power Running Board (PRB) motor. REFER to Power Running Board (PRB) Motor in this section.

DTC	Description	Action
B116A	RH Motor Traveled Too Far	RH motor is freewheeling (broken geartrain linkage). INSTALL a new RH PRB motor. REFER to Power Running Board (PRB) Motor in this section.
B116B	LH Power Running Board Motor Short	GO to Pinpoint Test D.
B116C	RH Power Running Board Motor Short	GO to Pinpoint Test D.
B116D	LH Power Running Board Motor Line Open	GO to Pinpoint Test D.
B116E	RH Power Running Board Motor Line Open	GO to Pinpoint Test D.
B1170	Right Power Running Board Hall Effect Sensor Failure	GO to Pinpoint Test E.
B1171	Left Power Running Board Hall Effect Signal Noisy	INSTALL a new LH PRB motor. REFER to Power Running Board (PRB) Motor in this section.
B1172	Right Power Running Board Hall Effect Signal Noisy	INSTALL a new RH PRB motor. REFER to Power Running Board (PRB) Motor in this section.
B1175	Right Power Running Board Stow Circuit Short To Ground	GO to Pinpoint Test D.
B1178	Right Power Running Board Deploy and/or Stow Circuit Short To Battery	GO to Pinpoint Test D.
B1317	Battery Voltage High	GO to Pinpoint Test F.
B1318	Battery Voltage Low	GO to Pinpoint Test F.
B1342	ECU is Faulted	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section.
B2139	Data Mismatch (receive data does not match what was expected)	REFER to Section 419-10.
B2813	LH Power Running Board Stow Time Fail	GO to Pinpoint Test B.
B2814	LH Power Running Board Deploy Time Fail	GO to Pinpoint Test B.
B2816	RH Power Running Board Stow Time Fail	GO to Pinpoint Test C.
B2817	RH Power Running Board Deploy Time Fail	GO to Pinpoint Test C.
U1900	CAN Communication Bus Fault - Receive Error	Do not install a new PRB module as part of the repair for a PRB DTC U1900 fault. The PRB is not the source of the fault when DTC U1900 is present. REFER to Section 418-00 to diagnose the Controller Area Network (CAN) concern.
U2050	No Application Present	INSTALL a new PRB module. REFER to Section 419-10. REPEAT the self-test.
All other DTCs	—	REFER to the Master DTC Chart in Section 419-10.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
The Power Running Board (PRB) does not operate correctly — both running boards	- Fuse - Circuitry - Smart Junction Box (SJB) - PRB module - PCM - Instrument Cluster (IC)	GO to Pinpoint Test A.
The PRB does not operate correctly — left running board does not deploy/stow or reverses direction	- Circuitry - Door ajar switch - PRB mechanical assembly - PRB motor - PRB module	GO to Pinpoint Test B.
The PRB does not operate correctly — right running board does not deploy/stow or reverses direction	- Circuitry - Door ajar switch - PRB mechanical assembly - PRB motor - PRB module	GO to Pinpoint Test C.

Pinpoint Tests

Pinpoint Test A: The Power Running Board (PRB) Does Not Operate Correctly — Both Running Boards

Refer to Wiring Diagrams Cell [109](#) , Power Liftgate/Retractable Running Boards for schematic and connector information.

NOTE: *The Power Running Board (PRB) module will stop functionality when battery voltage drops below 9.5 volts (early build) or 9.0 volts (late build). The PRB system will recover at the next door ajar transition provided that the battery voltage remains above 9.0 volts (late build only).*

Normal Operation

The PRB module receives power through circuit SBB11 (BU/RD) and ground through circuit GD134 (BK/VT). The LH and RH PRB motor Hall-effect sensors receive a regulated 10-volt power supply from the PRB module through circuit CPR47 (GY/BN). The Hall-effect ground is provided by the PRB module through circuit RPR47 (BU/GN). The PRB module receives door ajar and vehicle speed status over the Controller Area Network (CAN) .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- PCM
- Instrument Cluster (IC)
- ~~PRB~~ module

PINPOINT TEST A : THE POWER RUNNING BOARD (PRB) DOES NOT OPERATE CORRECTLY — BOTH RUNNING BOARDS

A1 RETRIEVE ALL CMDTCS IN ALL MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self-Test — ALL ~~CM~~DTCS .

Are any door ajar switch or vehicle speed related DTCs present?

Yes	REFER to Section 419-10 to diagnose the door ajar switch or vehicle speed related DTCs.
No	GO to A2 .

A2 CHECK THE OPERATION OF THE PRBS — AUTOMATIC MODE

- Set the ~~PRB~~ system to the AUTO (enabled) mode, where the boards automatically deploy when a door on the same side is opened, and retract when the door is closed.
- Open and close the LH front door. The LH ~~PRB~~ should deploy when the door is open, and retract when the door is closed.
- Open and close the LH rear door. The LH ~~PRB~~ should deploy when the door is open, and retract when the door is closed.
- Open and close the RH front door. The RH ~~PRB~~ should deploy when the door is open, and retract when the door is closed.
- Open and close the RH rear door. The RH ~~PRB~~ should deploy when the door is open, and retract when the door is closed.

Did both ~~PRBs~~ operate correctly, with no reversal of motion or noise while in motion?

Yes	The system is operating as designed. REVIEW operation of the PRB system with the customer.
No	GO to A3 .

A3 CHECK THE OPERATION OF THE PRBS — MANUAL MODE

- Set the ~~PRB~~ system to the OUT mode, where the boards are deployed all the time, regardless of door open/closed status and vehicle speed is less than 8 km/h (5 mph).

Did both ~~PRBs~~ deploy?

Yes	GO to A5 .
No	GO to A4 .

A4 CHECK THE PRBS FOR FOREIGN MATERIAL

- **NOTE:** *If the system has false bounce-back, stops in mid-travel, is noisy or will not deploy or stow completely, check to see if any foreign material is caught in the PRB hinge points, step area or door sill. If foreign material is found, clean the PRB hinge points, step area and door sill with a high-pressure washer. If the system operates correctly after cleaning, refer the customer to the Owner's Literature for correct cleaning instructions.*

Check the PRB hinge points, step area and door sill of the LH and RH PRBs for foreign material or blockage. If any foreign material or blockage is found, remove it, and if necessary, set the PRB system to the OUT mode and clean the PRB hinge points, step area and door sill with a high-pressure washer. Avoid direct spray to the motor.

- Set the PRB system to the AUTO (enabled) mode, where the boards automatically deploy when a door on the same side is opened, and retract when the door is closed.
- Open and close the LH front door. The LH PRB should deploy when the door is open, and retract when the door is closed.
- Open and close the RH front door. The RH PRB should deploy when the door is open, and retract when the door is closed.

Does the PRB system operate normally during all test steps?

Yes	The system is operating normally. The concern may have been caused by foreign material or blockage in the PRB hinge points, step area or door sill. REFER the customer to the Owner's Literature for correct cleaning instructions.
No	GO to A5 .

A5 VERIFY THE DOOR AJAR SWITCH OPERATION

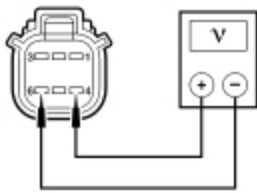
- Open and close each door, verify the interior lights operate correctly and door ajar is noted on the vehicle message center.

Do the interior lights turn ON when the door is opened and OFF when the door is closed?

Yes	GO to A6 .
No	REFER to Section 417-02 to diagnose the interior lighting concern before attempting to diagnose the PRB concern.

A6 CHECK THE HALL-EFFECT SENSOR POWER AND GROUND

- Disconnect: PRB LH Motor C3185 .
- Disconnect: PRB RH Motor C3186 .
- Ignition ON.
- Measure the voltage between RH PRB motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and [C3186](#) Pin 6, circuit RPR47 (BU/GN), harness side.



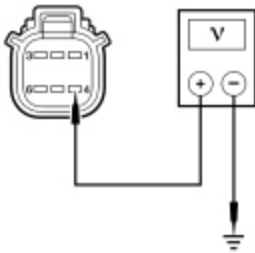
- N0092272
- Open RH door.
- Voltage signal will only be present for less than one second.

Is the voltage between 7 and 10.5 volts?

Yes	GO to A13 .
No	If the voltage was less than 7 volts, GO to A7 . If the voltage was greater than 10.5 volts, GO to A12 .

A7 CHECK HALL-EFFECT SENSOR POWER CIRCUIT FOR VOLTAGE

- Measure the voltage between RH ~~PRB~~ motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ground.



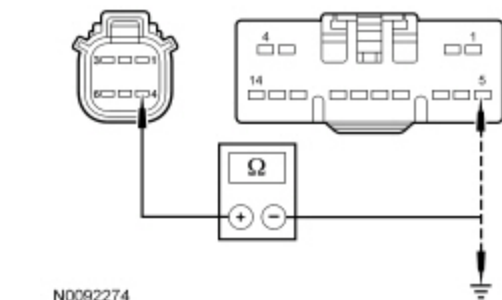
- N0092273
- Open RH door.
- Voltage signal will only be present for less than one second.

Is the voltage between 7 and 10.5 volts?

Yes	GO to A11 .
No	GO to A8 .

A8 CHECK HALL-EFFECT SENSOR POWER CIRCUIT FOR AN OPEN OR SHORT TO GROUND

- Ignition OFF.
- Disconnect: ~~PRB~~ Module C3313B .
- Measure the resistance between RH ~~PRB~~ motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ~~PRB~~ module [C3313B](#) Pin 5, circuit CPR47 (GY/BN), harness side; and between RH ~~PRB~~ motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ground.

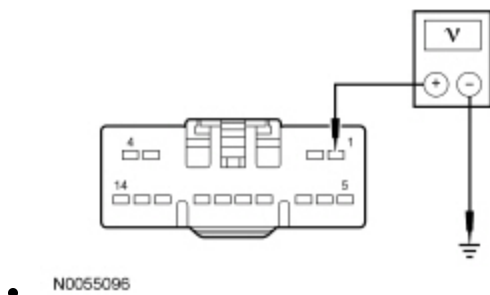


Is the resistance less than 5 ohms between the PRB motor and PRB module; and greater than 10,000 ohms between the PRB motor and ground?

Yes	GO to A9 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

A9 CHECK THE PRB MODULE POWER CIRCUIT FOR VOLTAGE

- Measure the voltage between PRB module [C3313B](#) Pin 1, circuit SBB11 (BU/RD), harness side and ground.

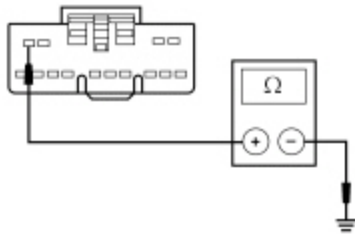


Is battery voltage present?

Yes	GO to A10 .
No	VERIFY Battery Junction Box (BJB) fuse 11 (30A) is OK. If OK, REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

A10 CHECK THE PRB MODULE GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the PRB module [C3313B](#) Pin 4, circuit GD139 (BK/YE), harness side and ground.



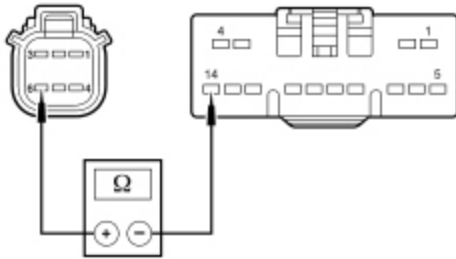
• N0060073

Is the resistance less than 5 ohms?

Yes	GO to A13 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

A11 CHECK THE HALL-EFFECT SENSOR GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: ~~PRB~~ Module C3313B .
- Measure the resistance between RH ~~PRB~~ motor [C3186](#) Pin 6, circuit RPR47 (BU/GN), harness side and ~~PRB~~ module [C3313B](#) Pin 14, circuit RPR47 (BU/GN), harness side.



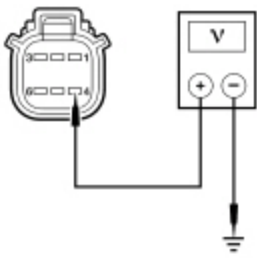
• N0092275

Is the resistance less than 5 ohms?

Yes	GO to A13 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

A12 CHECK THE HALL-EFFECT SENSOR POWER CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: ~~PRB~~ Module C3313B .
- Ignition ON.
- Measure the voltage between RH ~~PRB~~ motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ground.



• N0092273

Is any voltage present?

Yes	REPAIR the circuit. CLEAR The DTCs. REPEAT the self-test.
No	GO to A13 .

A13 CHECK THE PRB MODULE FOR CORRECT OPERATION

- Connect: ~~PRB~~ LH Motor C3185 .
- Connect: ~~PRB~~ RH Motor C3186 .
- Visually inspect all ~~PRB~~ module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all ~~PRB~~ module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test B: The Power Running Board (PRB) Does Not Operate Correctly — Left Running Board Does Not Deploy/Stow or Reverses Direction

Refer to Wiring Diagrams Cell [109](#) , Power Liftgate/Retractable Running Boards for schematic and connector information.

Normal Operation

During self-test, the Power Running Board (PRB) module monitors the length of time it takes to deploy and stow the LH and RH ~~PRBs~~ . If the time exceeds 4 seconds, DTC B2813 or B2814 will be generated.

The ~~PRB~~ module receives door ajar and vehicle speed status over the Controller Area Network (CAN) . During LH ~~PRB~~ deployment, the ~~PRB~~ module supplies power to the LH ~~PRB~~ motor through circuit CPR41 (BU/WH) and ground through

circuit CPR42 (WH). The PRB module stows the LH PRB by supplying power to the LH PRB motor through circuit CPR42 (WH) and ground through circuit CPR41 (BU/WH). The PRB module provides a regulated 10-volt power supply to the Hall-effect sensor (internal to the PRB motor) through circuit CPR47 (GY/BN). The Hall-effect ground is provided by the PRB module through circuit RPR47 (BU/GN). The PRB module monitors the LH PRB Hall-effect signal through circuit VPR43 (GY/BU). If the PRB module does not receive any signal from the Hall-effect sensor during operation, the PRB will stop.

- DTC B2813 (LH Power Running Board Stow Time Fail) — When the left running board takes greater than 4 seconds to stow.
- DTC B2814 (LH Power Running Board Deploy Time Fail) — When the left running board takes greater than 4 seconds to deploy.

This pinpoint test is intended to diagnose the following:

- Mechanical resistance due to foreign material
- Wiring, terminals or connectors
- Door ajar switch
- PRB module
- LH PRB motor

PINPOINT TEST B : THE POWER RUNNING BOARD (PRB) DOES NOT OPERATE CORRECTLY — LEFT RUNNING BOARD DOES NOT DEPLOY/STOW OR REVERSES DIRECTION

B1 CHECK THE OPERATION OF THE PRB — AUTOMATIC MODE

- Set the PRB system to the AUTO (enabled) mode, where the boards automatically deploy when a door on the same side is opened, and retract when the door is closed.
- Open and close the LH front door. The LH PRB should deploy when the door is open, and retract when the door is closed.
- Open and close the LH rear door. The LH PRB should deploy when the door is open, and retract when the door is closed.
- Open and close the RH front door. The RH PRB should deploy when the door is open, and retract when the door is closed.
- Open and close the RH rear door. The RH PRB should deploy when the door is open, and retract when the door is closed.

Did both PRBs operate correctly, with no reversal of motion, or noise while in motion?

Yes	The system is operating as designed. REVIEW operation of the PRBs system with the customer.
No	GO to B2 .

B2 CHECK THE OPERATION OF THE PRBS — MANUAL MODE

- Set the PRB system to the OUT mode, where the boards are deployed all the time, regardless of door open/closed status and vehicle speed is less than 8 km/h (5 mph).

Did both PRBs deploy?

Yes	GO to B5 .
No	GO to B3 .

B3 CHECK THE PRBS FOR FOREIGN MATERIAL

- **NOTE:** *If the system has false bounce-back, stops in mid-travel, is noisy or will not stow completely, extend the PRB and check to see if any foreign material is caught in the PRB hinge points, step area or door sill. If foreign material is found, clean the PRB hinge points, step area and door sill with a high-pressure washer. If the system operates correctly after cleaning, refer the customer to the Owner's Literature for correct cleaning instructions.*

Check the PRB hinge points, step area and door sill of the LH PRB for foreign material or blockage. If any foreign material or blockage is found, remove it, and if necessary, set the PRB system to the OUT mode and clean the PRB hinge points, step area and door sill with a high-pressure washer. Avoid direct spray to the motor.

- Set the PRB system to the AUTO (enabled) mode, where the boards automatically deploy when either door on the same side is opened, and retract when the door is closed.
- Open and close the LH front door. The LH PRB should deploy when the door is open, and retract when the door is closed.

Does the LH PRB operate normally during all test steps?

Yes	The system is operating normally. The concern may have been caused by foreign material or blockage in the PRB hinge points, step area or door sill. REFER the customer to the Owner's Literature for correct cleaning instructions.
No	GO to B4 .

B4 CHECK IF DTCS ARE PRESENT

- Use the recorded PRB module DTCs from the continuous and on-demand self tests.

Is DTC B2813 or B2814 present?

Yes	INSTALL a new PRB motor. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test.
No	GO to B6 .

B5 VERIFY THE DOOR AJAR SWITCH OPERATION

- Open and close the LH front and LH rear door, and verify the interior lights operate correctly and door ajar is noted on the vehicle message center.

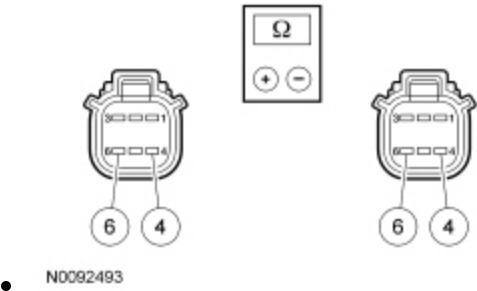
Do the interior lights turn ON when the door is opened, and OFF when the door is closed?

Yes	GO to B6 .
No	REFER to Section 417-02 to diagnose the interior lighting concern before attempting to diagnose the PRB concern.

B6 CHECK FOR CONTINUITY BETWEEN LH AND RH PRB MOTORS ON THE HALL-EFFECT SENSOR POWER AND GROUND CIRCUITS

- Disconnect: ~~PRB~~ LH Motor C3185 .
- Disconnect: ~~PRB~~ RH Motor C3186 .
- Ignition OFF.
- Measure the resistance between LH ~~PRB~~ motor [C3185](#), harness side and RH ~~PRB~~ motor [C3186](#), harness side using the following chart:

LH PRB Motor	Circuit	RH PRB Motor
C3185 Pin 4	CPR47 (GY/BN)	C3186 Pin 4
C3185 Pin 6	RPR47 (BU/GN)	C3186 Pin 6

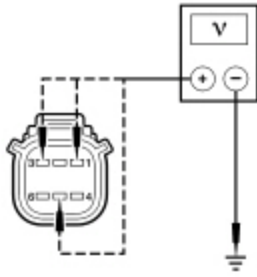


Is the resistance less than 5 ohms?

Yes	GO to B7 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

B7 CHECK FOR A SHORT TO VOLTAGE IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Ignition OFF.
- Disconnect: ~~PRB~~ Module C3313A .
- Disconnect: ~~PRB~~ Module C3313B .
- Ignition ON.
- Measure the voltage between LH ~~PRB~~ motor [C3185](#) Pin 5, circuit VPR43 (GY/BU), harness side and ground; and between LH ~~PRB~~ motor [C3185](#) Pin 3, circuit CPR41 (BU/WH), harness side and ground; and between [C3185](#) Pin 1, circuit CPR42 (WH), harness side and ground.



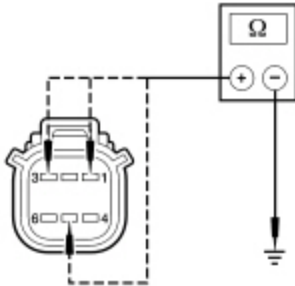
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Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
No	GO to B8 .

B8 CHECK FOR A SHORT TO GROUND IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Ignition OFF.
- Measure the resistance between LH ~~PRB~~ motor [C3185](#) Pin 5, circuit VPR43 (GY/BU), harness side and ground; and between LH ~~PRB~~ motor [C3185](#) Pin 3, circuit CPR41 (BU/WH), harness side and ground; and between [C3185](#) Pin 1, circuit CPR42 (WH), harness side and ground.



N0093022

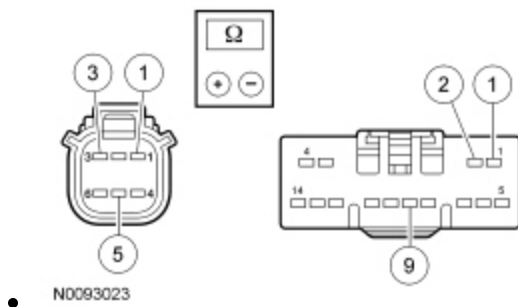
Is the resistance greater than 10,000 ohms?

Yes	GO to B9 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

B9 CHECK FOR AN OPEN IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Measure the resistances between ~~PRB~~ module [C3313A](#) or [C3313B](#), harness side and LH ~~PRB~~ motor [C3185](#), harness side using the following chart:

LH Motor Connector	Circuit	PRB Module Connector
C3185 Pin 5	VPR43 (GY/BU)	C3313B Pin 9
C3185 Pin 3	CPR41 (BU/WH)	C3313A Pin 1
C3185 Pin 1	CPR42 (WH)	C3313A Pin 2

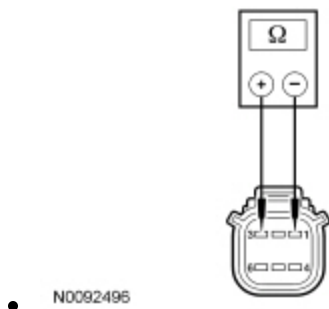


Is the resistance less than 5 ohms?

Yes	GO to B10 .
No	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test.

B10 CHECK FOR A SHORT BETWEEN THE STOW AND DEPLOY CIRCUITS

- Measure the resistance between LH PRB motor [C3185](#) Pin 3, circuit CPR41 (BU/WH), harness side and [C3185](#) Pin 1, circuit CPR42 (WH) harness side.



Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new LH PRB bracket assembly. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test. If the concern is still present, GO to B11 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

B11 CHECK THE PRB MODULE FOR CORRECT OPERATION

- Connect: PRB LH Motor C3185 .
- Connect: PRB RH Motor C3186 .
- Visually inspect all PRB module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all PRB module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test C: The Power Running Board (PRB) Does Not Operate Correctly — Right Running Board Does Not Deploy/Stow or Reverses Direction

Refer to Wiring Diagrams Cell 109 , Power Liftgate/Power Running Boards for schematic and connector information.

Normal Operation

During self-test, the Power Running Board (PRB) module monitors the length of time it takes to deploy and stow the LH and RH PRBs . If the time exceeds 4 seconds, B2816 or B2817 will be generated.

The PRB module receives door ajar and vehicle speed status over the Controller Area Network (CAN) . During RH PRB deployment, the PRB module supplies power to the RH PRB motor through circuit CPR44 (BN/WH) and ground through circuit CPR45 (VT/GY). The PRB module stows the RH PRB by supplying power to the RH PRB motor through circuit CPR45 (VT/GY) and ground through circuit CPR44 (BN/WH). The PRB module provides a regulated 10-volt power supply to the Hall-effect sensor (internal to the PRB motor) through circuit CPR47 (GY/BN). The Hall-effect ground is provided by the PRB module through circuit RPR47 (BU/GN). The PRB module monitors the RH PRB Hall-effect signal through circuit VPR46 (GY). If the PRB module does not receive any signal from the Hall-effect sensor during operation, the PRB will stop.

- DTC B2816 (RH Power Running Board Stow Time Fail) — Right running board takes greater than 4 seconds to stow.
- DTC B2817 (RH Power Running Board Deploy Time Fail) — Right running board takes greater than 4 seconds to deploy.

This pinpoint test is intended to diagnose the following:

- Mechanical resistance due to foreign material
- Wiring, terminals or connectors
- Door ajar switch
- PRB module
- RH PRB motor

PINPOINT TEST C : THE POWER RUNNING BOARD (PRB) DOES NOT OPERATE CORRECTLY — RIGHT RUNNING BOARD DOES NOT DEPLOY/STOW OR REVERSES DIRECTION

C1 CHECK THE OPERATION OF THE PRB — AUTOMATIC MODE
• Set the PRB system to the AUTO (enabled) mode, where the boards automatically deploy when a door on the same side is opened, and retract when the door is closed. • Open and close the LH front door. The LH PRB should deploy when the door is open, and retract when the door is closed. • Open and close the LH rear door. The LH PRB should deploy when the door is open, and retract when the door is closed. • Open and close the RH front door. The RH PRB should deploy when the door is open, and retract when the door is closed.

- Open and close the RH rear door. The RH PRB should deploy when the door is open, and retract when the door is closed.

Did both PRBs operate correctly, with no reversal of motion or noise while in motion?

Yes	The system is operating as designed. REVIEW operation of the PRBs system with the customer.
No	GO to C2 .

C2 CHECK THE OPERATION OF THE PRBS — MANUAL MODE

- Set the PRB system to the OUT mode, where the boards are deployed all the time, regardless of door open/closed status and vehicle speed is less than 8 km/h (5 mph).

Did both PRBs deploy?

Yes	GO to C5 .
No	GO to C3 .

C3 CHECK THE PRB FOR FOREIGN MATERIAL

- **NOTE:** *If the system has false bounce-back, stops in mid-travel, is noisy or will not stow completely, extend the PRB and check to see if any foreign material is caught in the PRB hinge points, step area or door sill. If foreign material is found, clean the PRB hinge points, step area and door sill with a high-pressure washer. If the system operates correctly after cleaning, refer the customer to the Owner's Literature for correct cleaning instructions.*

Check the PRB hinge points, step area and door sill of the RH PRB for foreign material or blockage. If any foreign material or blockage is found, remove it, and if necessary, set the PRB system to the OUT mode and clean the PRB hinge points, step area and door sill with a high-pressure washer. Avoid direct spray to the motor.

- Set the PRB system to the AUTO (enabled) mode, where the boards automatically deploy when either door on the same side is opened, and retract when the door is closed.
- Open and close the RH front door. The RH PRB should deploy when the door is open, and retract when the door is closed.

Does the RH PRB operate normally during all test steps?

Yes	The system is operating normally. The concern may have been caused by foreign material or blockage in the PRB hinge points, step area or door sill. REFER the customer to the Owner's Literature for correct cleaning instructions.
No	GO to C4 .

C4 CHECK IF DTCS ARE PRESENT

- Use the recorded PRB module DTCs from the continuous and on-demand self tests.

Is DTC B2816 or B2817 present?

Yes	INSTALL a new PRB motor. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test.
No	GO to C6 .

C5 VERIFY THE DOOR AJAR SWITCH OPERATION

- Open and close the RH front and RH rear door, and verify the interior lights operate correctly and door ajar is noted on the vehicle message center.

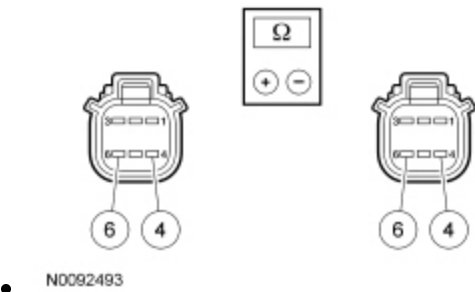
Do the interior lights turn ON when the door is opened, and OFF when the door is closed?

Yes	GO to C6 .
No	REFER to Section 417-02 to diagnose the interior lighting concern before attempting to diagnose the PRB concern.

C6 CHECK FOR CONTINUITY BETWEEN LH AND RH PRB MOTORS ON THE HALL-EFFECT SENSOR POWER AND GROUND CIRCUITS

- Disconnect: **PRB** LH Motor C3185 .
- Disconnect: **PRB** RH Motor C3186 .
- Ignition OFF.
- Measure the resistance between LH **PRB** motor [C3185](#), harness side and RH **PRB** motor [C3186](#), harness side using the following chart:

LH PRB Motor	Circuit	RH PRB Motor
C3185 Pin 4	CPR47 (GY/BN)	C3186 Pin 4
C3185 Pin 6	RPR47 (BU/GN)	C3186 Pin 6

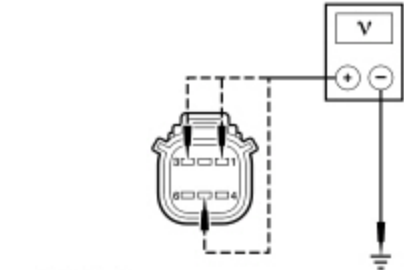


Is the resistance less than 5 ohms?

Yes	GO to C7 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

C7 CHECK FOR A SHORT TO VOLTAGE IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Ignition OFF.
- Disconnect: ~~PRB~~ Module C3313A .
- Disconnect: ~~PRB~~ Module C3313B .
- Ignition ON.
- Measure the voltage between RH ~~PRB~~ motor [C3186](#) Pin 5, circuit VPR46 (GY), harness side and ground; and between RH ~~PRB~~ motor [C3186](#) Pin 3, circuit CPR44 (BN/WH), harness side and ground; and between [C3186](#) Pin 1, circuit CPR45 (VT/GY), harness side and ground.



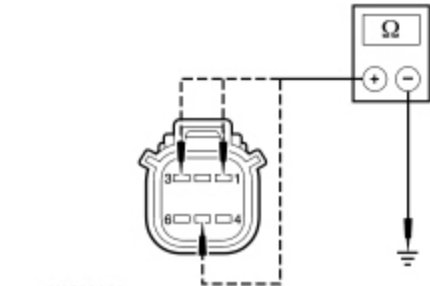
• N0092494

Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
No	GO to C8 .

C8 CHECK FOR A SHORT TO GROUND IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Ignition OFF.
- Measure the resistance between RH ~~PRB~~ motor [C3186](#) Pin 5, circuit VPR46 (GY), harness side and ground; and between RH ~~PRB~~ motor [C3186](#) Pin 3, circuit CPR44 (BN/WH), harness side and ground; and between [C3186](#) Pin 1, circuit CPR45 (VT/GY), harness side and ground.



• N0093022

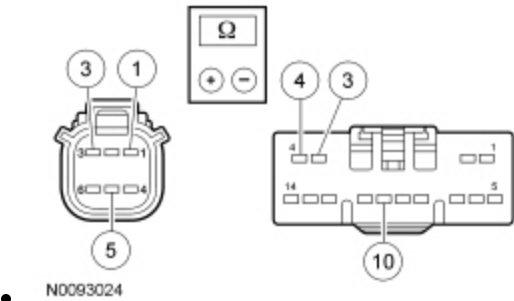
Is the resistance greater than 10,000 ohms?

Yes	GO to C9 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

C9 CHECK FOR AN OPEN IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Measure the resistances between PRB module C3313A or C3313B, harness side and RH PRB motor C3185, harness side using the following chart:

RH Motor Connector	Circuit	PRB Module Connector
C3186 Pin 5	VPR46 (GY)	C3313B Pin 10
C3186 Pin 3	CPR44 (BN/WH)	C3313A Pin 4
C3186 Pin 1	CPR45 (VT/GY)	C3313A Pin 3

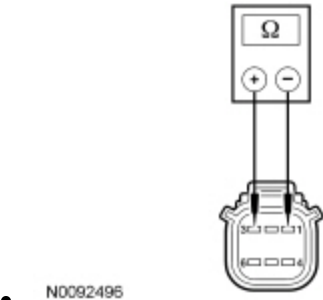


Is the resistance less than 5 ohms?

Yes	GO to C10.
No	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test.

C10 CHECK FOR A SHORT BETWEEN THE STOW AND DEPLOY CIRCUITS

- Measure the resistance between RH PRB motor C3186 Pin 3, circuit CPR44 (BN/WH), harness side and C3186 Pin 1, circuit CPR45 (VT/GY), harness side.



Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new RH PRB bracket assembly. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test. If the concern is still present, GO to C11.
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

C11 CHECK THE PRB MODULE FOR CORRECT OPERATION

- Connect: PRB LH Motor C3185 .
- Connect: PRB RH Motor C3186 .
- Visually inspect all PRB module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all PRB module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test D: DTCs B1167, B1168, B1175, B1178, B116B, B116D, B116C and B116E

Refer to Wiring Diagrams Cell [109](#) , Power Liftgate/Power Running Boards for schematic and connector information.

Normal Operation

During Power Running Board (PRB) deployment, the PRB module supplies power to the PRB motor through circuit CPR41 (BU/WH) (LH) or CPR44 (BN/WH) (RH) and ground through circuit CPR42 (WH) (LH) or CPR45 (VT/GY) (RH). The PRB module stows the PRB by supplying power to the PRB motor through circuit CPR42 (WH) (LH) or CPR45 (VT/GY) (RH) and ground through circuit CPR41 (BU/WH) (LH) or CPR44 (BN/WH) (RH). The PRB module provides a regulated 10-volt power supply to the Hall-effect sensor (internal to the PRB motor) through circuit CPR47 (GY/BN). The Hall-effect ground is provided by the PRB module through circuit RPR47 (BU/GN). The PRB module monitors the PRB Hall-effect signal through circuit VPR43 (GY/BU) (LH) or VPR46 (GY) (RH). If the PRB module does not receive any signal from the Hall-effect sensor during operation, the PRB will stop.

DTC Description	Fault Trigger Conditions
• B1167 — Left Power Running Board Deploy and/or Stow Circuit Short to Battery	The PRB module sets this DTC if a short to battery voltage is detected in the LH running board motor circuit(s).
• B1168 — Left Power Running Board Motor Circuit Short to Ground	The PRB module sets this DTC if a short to ground is detected in the motor circuit while attempting to retract/stow the LH running board.

DTC Description	Fault Trigger Conditions
B1175 — Right Power Running Board Stow Circuit Short to Ground	The PRB module sets this DTC if a short to ground is detected in the motor circuit while attempting to retract/stow the RH running board.
B1178 — Right Power Running Board Deploy and/or Stow Circuit Short to Battery	The PRB module sets this DTC if a short to battery voltage is detected in the RH running board motor circuit(s).
B116B — LH Power Running Board Motor Short	The PRB module sets this DTC if it detects the LH running board motor circuit(s) are shorted together.
B116C — RH Power Running Board Motor Short	The PRB module sets this DTC if it detects the RH running board motor circuit(s) are shorted together.
B116D — LH Power Running Board Motor Line Open	The PRB module sets this DTC when it detects an open in the LH running board motor circuit(s).
B116E — RH Power Running Board Motor Line Open	The PRB module sets this DTC when it detects an open in the RH running board motor circuit(s).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- LH or RH **PRB** motor assembly
- **PRB** module

PINPOINT TEST D : DTCS B1167, B1168, B1175, B1178, B116B, B116D, B116C AND B116E

D1 CHECK FOR A SHORT TO VOLTAGE IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT
• Ignition OFF. • Disconnect: PRB Module C3313A . • Disconnect: PRB Module C3313B . • Ignition ON. • For DTC B1167, B1168, B116B or B116D, measure the voltage between LH PRB motor C3185 Pin 5, circuit VPR43 (GY), harness side and ground; and between LH PRB motor C3185 Pin 3, circuit CPR41 (BU/WH), harness side and ground; and between C3185 Pin 1, circuit CPR42 (WH), harness side and ground. • N0092494 • For DTC B1175, B1178, B116C or B116E, measure the voltage between RH PRB motor C3186 Pin 5, circuit VPR46 (GY/OG), harness side and ground; and between RH PRB motor C3186 Pin 3, circuit CPR44

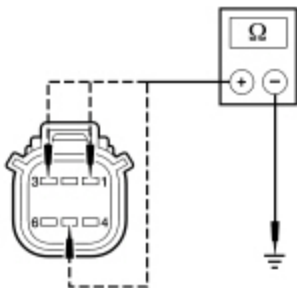
(BN/WH), harness side and ground; and between [C3186](#) Pin 1, circuit CPR45 (VT/GY), harness side and ground.

Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
No	GO to D2 .

D2 CHECK FOR A SHORT TO GROUND IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Ignition OFF.
- For DTC B1167, B1168, B116B or B116D, measure the resistance between LH ~~PRB~~ motor [C3185](#) Pin 5, circuit VPR43 (GY/BU), harness side and ground; and between LH ~~PRB~~ motor [C3185](#) Pin 3, circuit CPR41 (BU/WH), harness side and ground; and between [C3185](#) Pin 1, circuit CPR42 (WH), harness side and ground.



- N0093022
- For DTC B1175, B1178, B116C or B116E, measure the resistance between RH ~~PRB~~ motor [C3186](#) Pin 5, circuit VPR46 (GY), harness side and ground; and between RH ~~PRB~~ motor [C3186](#) Pin 3, circuit CPR44 (BN/WH), harness side and ground; and between [C3186](#) Pin 1, circuit CPR45 (VT/GY), harness side and ground.

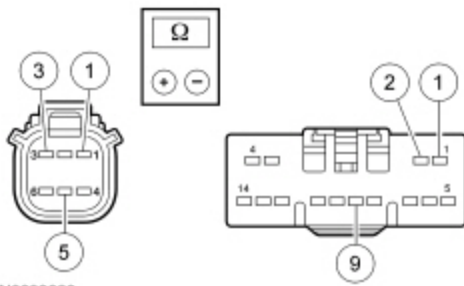
Is the resistance greater than 10,000 ohms?

Yes	GO to D3 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

D3 CHECK FOR AN OPEN IN THE HALL-EFFECT SENSOR SIGNAL, DEPLOY OR STOW CIRCUIT

- Ignition OFF.
- For DTC B1167, B1168, B116B or B116D, measure the resistances between ~~PRB~~ module [C4322A](#) or [C4322B](#), harness side and LH ~~PRB~~ motor [C3185](#), harness side using the following chart:

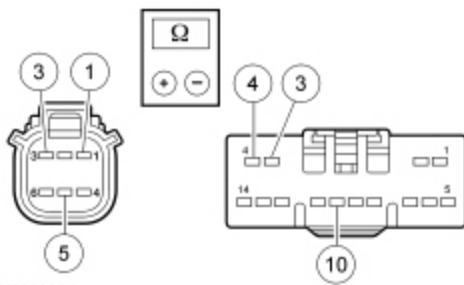
LH Motor Connector	Circuit	PRB Module Connector
C3185 Pin 5	VPR43 (GY/BU)	C3313B Pin 9
C3185 Pin 3	CPR41 (BU/WH)	C3313A Pin 1
C3185 Pin 1	CPR42 (WH)	C3313A Pin 2



N0093023

- For DTC B1175, B1178, B116C or B116E, measure the resistances between PRB module [C4322A](#) or [C4322B](#), harness side and RH PRB motor [C3186](#), harness side using the following chart:

RH Motor Connector	Circuit	PRB Module Connector
C3186 Pin 5	VPR46 (GY)	C3313B Pin 10
C3186 Pin 3	CPR44 (BN/WH)	C3313A Pin 4
C3186 Pin 1	CPR45 (VT/GY)	C3313A Pin 3



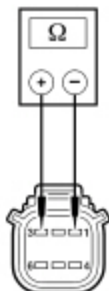
N0093024

Is the resistance less than 5 ohms?

Yes	GO to D4 .
No	REPAIR the circuit(s). CLEAR the DTCs. REPEAT the self-test.

D4 CHECK FOR A SHORT BETWEEN THE STOW AND DEPLOY CIRCUITS

- For DTC B1167, B1168, B116B or B116D, measure the resistance between LH PRB motor [C3185](#) Pin 3, circuit CPR41 (BU/WH), harness side and [C3185](#) Pin 1, circuit CPR42 (WH), harness side.



N0092495

- For DTC B1175, B1178, B116C or B116E, measure the resistance between RH PRB motor [C3186](#) Pin 3, circuit CPR44 (BN/WH), harness side and [C3186](#) Pin 1, circuit CPR45 (VT/GY), harness side.

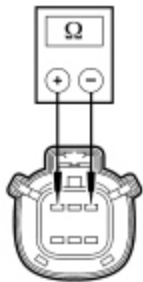
Is the resistance greater than 10,000 ohms?

Yes	GO to D5 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

D5 CHECK THE PRB MOTOR FOR AN OPEN CIRCUIT

NOTE: The running board motor(s) has approximately 0.42 ohms of resistance at 20°C (68°F). The running board motor(s) should not show an open circuit (no continuity) when measured with a DMM .

- For DTC B1167, B1168, B116B or B116D, measure the resistance between LH PRB motor [C3185](#) Pin 3, circuit CPR41 (BU/WH), component side and [C3185](#) Pin 1, circuit CPR42 (WH), component side.



• N0092495

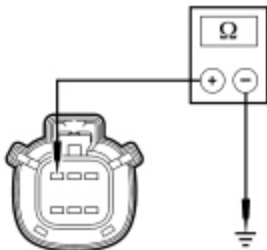
- For DTC B1175, B1178, B116C or B116E, measure the resistance between RH PRB motor [C3186](#) Pin 3, circuit CPR44 (BN/WH), component side and [C3186](#) Pin 1, circuit CPR45 (VT/GY), component side.

Is any continuity present?

Yes	GO to D6 .
No	INSTALL a new PRB motor. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test.

D6 CHECK THE PRB MOTOR FOR A SHORT TO GROUND

- For DTC B1167, B1168, B116B or B116D, measure the resistance between LH PRB motor [C3185](#) Pin 1, circuit CPR 42 (WH), component side and ground.



• N0092497

- For DTC B1175, B1178, B116C or B116E, measure the resistance between RH motor [C3186](#) Pin 1, circuit CPR45 (VT/GY), component side and ground.

Is the resistance greater than 10,000 ohms?

Yes	GO to D7 .
No	INSTALL a new PRB motor. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test.

D7 CHECK THE PRB MODULE FOR CORRECT OPERATION

- Connect: **PRB** LH Motor C3185 .
- Connect: **PRB** RH Motor C3186 .
- Visually inspect all **PRB** module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all **PRB** module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test E: DTCs B1162, B1163, B1164 and B1170

Refer to Wiring Diagrams Cell [109](#) , Power Liftgate/Power Running Boards for schematic and connector information.

Normal Operation

The Power Running Board (PRB) module provides a regulated 10-volt power supply to the Hall-effect sensor (internal to the **PRB** motor) through circuit CPR47 (GY/BN). The Hall-effect ground is provided by the **PRB** module through circuit RPR47 (BU/GN). The **PRB** module monitors the **PRB** Hall-effect signal through circuit VPR43 (GY/BU) (LH) or VPR46 (GY) (RH). If the **PRB** module does not receive any signal from the Hall-effect sensor during operation, the **PRB** will stop.

DTC Description	Fault Trigger Conditions
• B1162 — Power Running Board Hall-Effect Sensor Power Circuit Short to Battery	The PRB module sets this DTC if it detects the Hall-Effect sensor power circuit is shorted to battery voltage.
• B1163 — Left Power Running Board Hall-Effect Sensor Failure	The PRB module sets this DTC if it does not see any feedback from the Hall-Effect sensor during the stowing/deploying of the LH running board.

DTC Description	Fault Trigger Conditions
B1164 — Power Running Board Hall-Effect Sensor Power Circuit Short to Ground	The PRB module sets this DTC if it detects the Hall-Effect sensor power circuit is shorted to ground.
B1170 — Right Power Running Board Deploy and/or Stow Circuit Short to Battery	The PRB module sets this DTC if it does not see any feedback from the Hall-Effect sensor during the stowing/deploying of the RH running board.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- **PRB** module
- LH or RH **PRB** motor assembly

PINPOINT TEST E : DTCS B1162, B1163, B1164 AND B1170

E1 CHECK FOR CONTINUITY BETWEEN LH AND RH PRB MOTORS ON THE HALL-EFFECT SENSOR POWER AND GROUND CIRCUITS

• Disconnect: ~~PRB~~ LH Motor C3185 .

• Disconnect: ~~PRB~~ RH Motor C3186 .

• Ignition OFF.

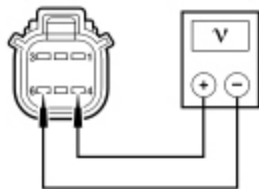
• Measure the resistance between LH ~~PRB~~ motor [C3185](#), harness side and RH ~~PRB~~ motor [C3186](#), harness side using the following chart:

LH PRB Motor	Circuit	RH PRB Motor
C3185 Pin 4	CPR47 (GY/BN)	C3186 Pin 4
C3185 Pin 6	RPR47 (BU/GN)	C3186 Pin 6

<

E2 CHECK THE HALL-EFFECT SENSOR POWER AND GROUND

- Ignition ON.
- Measure the voltage between RH PRB motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and [C3186](#) Pin 6, circuit RPR47 (BU/GN), harness side.



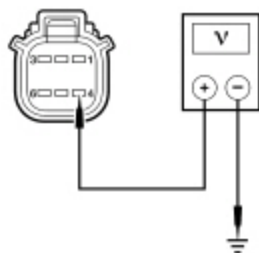
- N0092272
- Open RH door.
- Voltage signal will only be present for less than one second.

Is the voltage between 7 and 10.5 volts?

Yes	GO to E7 .
No	If the voltage was less than 7 volts, GO to E3 . If the voltage was greater than 10.5 volts, GO to E6 .

E3 CHECK HALL-EFFECT SENSOR POWER CIRCUIT FOR VOLTAGE

- Measure the voltage between RH PRB motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ground.



- N0092273
- Open RH door.
- Voltage signal will only be present for less than one second.

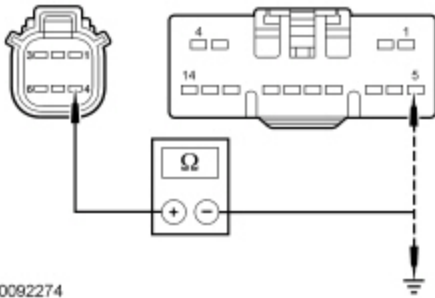
Is the voltage between 7 and 10.5 volts?

Yes	GO to E5 .
No	GO to E4 .

E4 CHECK HALL-EFFECT SENSOR POWER CIRCUIT FOR AN OPEN OR SHORT TO GROUND

- Ignition OFF.

- Disconnect: **PRB** Module C3313B .
- Measure the resistance between RH **PRB** motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and **PRB** module [C3313B](#) Pin 5, circuit CPR47 (GY/BN), harness side; and between RH **PRB** motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ground.

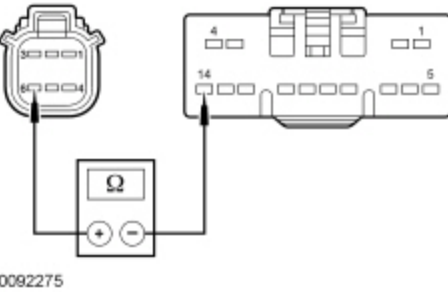


Is the resistance less than 5 ohms between the **PRB motor and the **PRB** module; and greater than 10,000 ohms between the **PRB** motor and ground?**

Yes	GO to E9 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

E5 CHECK THE HALL-EFFECT SENSOR GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: **PRB** Module C3313B .
- Measure the resistance between RH **PRB** motor [C3186](#) Pin 6, circuit RPR47 (BU/GN), harness side and **PRB** module [C3313B](#) Pin 14, circuit RPR47 (BU/GN), harness side.



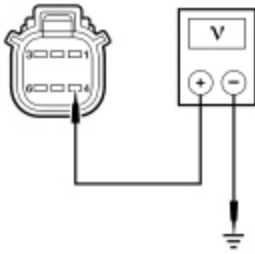
Is the resistance less than 5 ohms?

Yes	GO to E9 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

E6 CHECK THE HALL-EFFECT SENSOR POWER CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: **PRB** Module C3313B .
- Ignition ON.

- Measure the voltage between RH **PRB** motor [C3186](#) Pin 4, circuit CPR47 (GY/BN), harness side and ground.



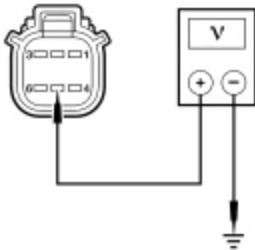
• N0092273

Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
No	GO to E9 .

E7 CHECK THE HALL-EFFECT SENSOR SIGNAL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: **PRB** Module C3313B .
- Ignition ON.
- For DTC B1163, measure the voltage between LH **PRB** motor [C3185](#) Pin 5, circuit VPR43 (GY/BU), harness side and ground.



• N0092498

- For DTC B1170, measure the voltage between RH **PRB** motor [C3186](#) Pin 5, circuit VPR46 (GY), harness side and ground.

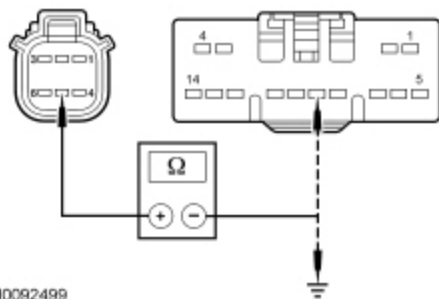
Is any voltage present?

Yes	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.
No	GO to E8 .

E8 CHECK THE HALL-EFFECT SENSOR SIGNAL CIRCUIT FOR AN OPEN OR SHORT TO GROUND

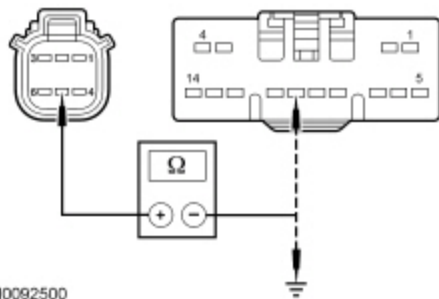
- Ignition OFF.
- For DTC B1163, measure the resistance between LH **PRB** motor [C3185](#) Pin 5, circuit VPR43 (GY/BU), harness side and **PRB** module [C3313B](#) Pin 9, circuit VPR43 (GY/BU), harness side; and between LH **PRB**

motor [C3185](#) Pin 5, circuit VPR43 (GY/BU), harness side and ground.



• N0092499

- For DTC B1170, measure the resistance between RH ~~PRB~~ motor [C3186](#) Pin 5, circuit VPR46 (GY), harness side and ~~PRB~~ module [C3313B](#) Pin 10, circuit VPR46 (GY), harness side; and between RH ~~PRB~~ motor [C3186](#) Pin 5, circuit VPR46 (GY), harness side and ground.



• N0092500

Is the resistance less than 5 ohms between the ~~PRB~~ motor and ~~PRB~~ module; and greater than 10,000 ohms between the ~~PRB~~ motor and ground?

Yes	GO to E10 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the self-test.

E9 CHECK THE PRB MODULE FOR CORRECT OPERATION

- Connect: ~~PRB~~ LH Motor C3185 .
- Connect: ~~PRB~~ RH Motor C3186 .
- Disconnect all ~~PRB~~ module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all ~~PRB~~ module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section. CLEAR the DTCs. REPEAT the self-test.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

E10 CHECK THE PRB MODULE FOR CORRECT OPERATION

- Connect: **PRB** LH Motor C3185 .
- Connect: **PRB** RH Motor C3186 .
- Disconnect all **PRB** module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all **PRB** module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	GO to E11 .
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

E11 CHECK THE PRB FOR FOREIGN MATERIAL

- **NOTE:** *If the system has false bounce-back, stops in mid-travel, is noisy or will not deploy or stow completely, extend the **PRB** and check to see if any foreign material is caught in the **PRB** hinge points, step area or door sill. If foreign material is found, clean the **PRB** hinge points, step area and door sill with a high-pressure washer. If the system operates correctly after cleaning, refer the customer to the Owner's Literature for correct cleaning instructions.*

Check the **PRB** hinge points, step area and door sill of the **PRBs** for foreign material or blockage. If any foreign material or blockage is found, remove it, and if necessary, set the **PRB** system to the OUT mode and clean the **PRB** hinge points, step area and door sill with a high-pressure washer. Avoid direct spray to the motor.

- Set the **PRB** system to the AUTO (enabled) mode, where the boards automatically deploy when either door on the same side is opened, and retract when the door is closed.
- Open and close the LH front door. The LH **PRB** should deploy when the door is open, and retract when the door is closed.
- Open and close the RH front door. The RH **PRB** should deploy when the door is open, and retract when the door is closed.

Do the LH and RH **PRBs operate normally during all test steps?**

Yes	The system is operating normally. The concern may have been caused by foreign material or blockage in the pivot areas or track guides. REFER the customer to the Owner's Literature for correct cleaning instructions.
No	INSTALL a new PRB motor. REFER to Power Running Board (PRB) Motor in this section. CLEAR the DTCs. REPEAT the self-test.

Pinpoint Test F: DTC B1317/B1318 — Battery Voltage High/Battery Voltage Low

Refer to Wiring Diagrams Cell [109](#) , Power Liftgate/Retractable Running Boards for schematic and connector information.

NOTE: *DTC B1317 or B1318 can be set if the vehicle has been recently jump started, the battery has been recently charged or the battery has been discharged. The battery may become discharged due to excessive load(s) on the charging system from aftermarket accessories or if the battery has been left unattended with the accessories on.*

Normal Operation

The Power Running Board (PRB) module continuously monitors the input voltage for correct operation. If the PRB module detects input voltage below 9.5 volts, it will store DTC B1318 in memory. If the PRB module detects input voltage greater than 16 volts, it will store DTC B1317 in memory. If the PRB module sets DTC B1317 or B1318, it will inhibit the power liftgate system operation.

- DTC B1317 (Battery Voltage High) — If the PRB module detects voltage above 16 volts while the running boards are in motion (deploy/stow), it will set this DTC. Once this DTC is set, the PRB system will be disabled until the battery voltage returns to normal.
- DTC B1318 (Battery Voltage Low) — If the PRB module detects voltage below 9.5 volts while the running boards are in motion (deploy/stow), it will set this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PRB module

PINPOINT TEST F : DTC B1317/B1318 — BATTERY VOLTAGE HIGH/BATTERY VOLTAGE LOW

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*

F1 RETRIEVE ALL CMDTCS IN ALL MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ALL CMDTCS .

Is DTC B1317 or B1318 retrieved from one or more modules and P0563, P0620, P0626 or P065B retrieved from the PCM?

Yes	REFER to Section 414-00 to diagnose the charging system. CLEAR all Continuous Memory Diagnostic Trouble Codes (CMDTCS) . REPEAT the PRB module self-test.
No	GO to F2 .

F2 TEST BATTERY CONDITION

- Ignition OFF.
- Carry out the Battery Condition Test. Refer to Section 414-01.

Did the battery pass the condition test?

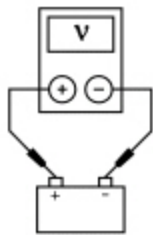
Yes	If the battery passed the condition test but required a recharge, REFER to Section 414-00 to diagnose the charging system. CLEAR all CMDTCS . REPEAT the PRB module self-test.
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	If the battery passed the condition test and did not require a recharge, GO to F3 .
No	INSTALL a new battery. CLEAR all <u>CMDTCs</u> . REPEAT the <u>PRB</u> module self-test.

F3 CHECK CHARGING SYSTEM VOLTAGE

NOTE: Do not allow the engine speed to increase above 2,000 rpm while performing this step or the generator may self excite and result in default charging system output voltage. If engine speed goes above 2,000 rpm, shut the vehicle OFF and restart the engine before performing this step.

- Measure the voltage of the battery:
 - For DTC B1317, turn off all accessories and run the engine at 1,500 rpm for a minimum of 2 minutes while measuring battery voltage.
 - For DTC B1318, turn on headlights and HVAC fan on high and run engine at 1,500 rpm for a minimum of 2 minutes while measuring battery voltage.



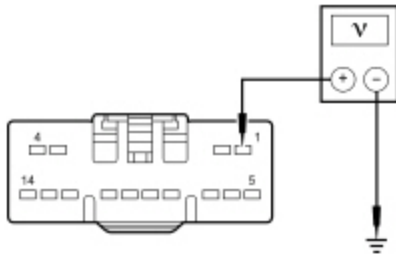
AJ0210-A

Is the voltage between 13 and 15.2 volts?

Yes	For DTC B1318, GO to F4 . For DTC B1317, GO to F6 .
No	REFER to Section 414-00 to diagnose the charging system. CLEAR all <u>CMDTCs</u> . REPEAT the <u>PRB</u> module self-test.

F4 CHECK THE PRB MODULE POWER CIRCUIT

- Ignition OFF.
- Disconnect: PRB module C3313B .
- Measure the voltage between PRB module [C3313B](#) Pin 1, circuit SBB11 (BU/RD), harness side and ground.



N0055095

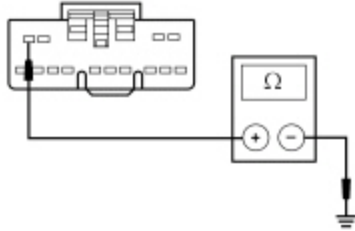
Is the voltage greater than 10 volts?

Yes	GO to F5 .
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No	VERIFY Battery Junction Box (BJB) fuse 11 (30A) is OK. If OK, REPAIR the circuit. CLEAR the DTCs. REPEAT the PRB module self-test. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. CLEAR the DTCs. REPEAT the PRB module self-test.
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F5 CHECK THE PRB MODULE GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between **PRB** module [C3313B](#) Pin 4, circuit GD139 (BK/YE), harness side and ground.



- N0060073

Is the resistance less than 5 ohms?

Yes	GO to F6 .
No	REPAIR the circuit. CLEAR the DTCs. REPEAT the PRB module self-test.

F6 CHECK FOR PRB MODULE OPERATION

- Check **PRB** module C3313A and C3313B for the following:
 - corrosion.
 - damaged pins.
 - pushed-out pins.
- Connect **PRB** module C3313A and C3313B and make sure the connectors seat correctly.
- Clear all **CM**DTCs .
- Operate the system to verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new PRB module. REFER to Power Running Board (PRB) Module in this section. REPEAT the PRB module self-test.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the PRB module self-test.